DATAWISE AI

DataWise AI: Research & Roadmap

Empirical Study Protocols, Comparative Benchmarks & Future Architectural Horizons

Document Type:	Scientific Research & Product Roadmap
Target Audience:	Research Collaborators, Academic Evaluators, Strategic Partners
Document Version:	v1.0.0 (Engineering Release)
Date of Issue:	October 27, 2026
Repository Status:	Verified Codebase & Production Architecture
Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

1. Research Questions & Scientific Hypotheses

DataWise AI represents a novel synthesis of LLM-based agent reasoning and deterministic machine learning pipelines. We formalize four primary research questions for ongoing empirical investigation:

- **RQ1 (Efficiency):** Does multi-agent adaptive candidate selection reduce total GPU/CPU compute cycles compared to exhaustive grid searches across Auto-sklearn and FLAML?
- **RQ2 (Leakage Immunity):** Does explicit graph-level leakage verification eliminate test set contamination across complex multi-table and temporal datasets?
- **RQ3 (Usability & Intent):** How effectively does the two-prompt onboarding protocol capture business constraints compared to unstructured natural language prompts?
- **RQ4 (Resilience):** What is the empirical fault-recovery rate of the 18-module resilience subsystem under simulated API rate limits and network degradation?

2. Comparative Architectural Benchmark

Capability Dimension	DataWise AI	Auto-sklearn	FLAML	AutoGluon
Reasoning Architecture	LangGraph 16 Agents	SMAC Optimization	Cost-Frugal Search	Multi-layer Stacking
Conversational Interface	Two-Prompt + Chat	None (CLI/API)	None (Python API)	None (Python API)
Data Leakage Auditing	Explicit Dedicated Agent	Implicit Pipeline	Implicit Pipeline	Internal Partitioning
Explainable Diagnostics	Full Text + Charts	Metric Table Only	Metric Table Only	Feature Importances
Artifact Bundling	pkl + ipynb + PDF + API	Pickle Model Only	Pickle Model Only	Pickle Directory
Resilience Subsystem	18 Recovery Modules	Standard Exceptions	Standard Exceptions	Standard Exceptions

3. Future Engineering & Research Roadmap

- **Phase 1 (Q1 2027):** Automated feature store synchronization (Feast integration) and online streaming inference via Kafka.
- **Phase 2 (Q2 2027):** Multi-modal dataset intelligence supporting mixed tabular, unstructured text, and image embeddings.
- **Phase 3 (Q3 2027):** Distributed multi-node Ray integration for parallelized hyperparameter exploration across cloud clusters.